

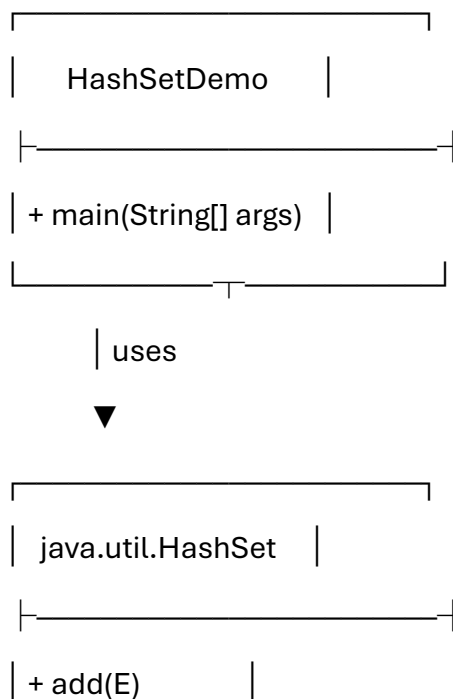
Aim

To write a Java program to demonstrate the important methods of the **HashSet class**, such as `add()`, `remove()`, `contains()`, `size()`, `isEmpty()`, and `clear()`.

Algorithm

1. Start the program.
2. Import the `java.util.HashSet` class.
3. Create a `HashSet` object.
4. Use `add()` to insert elements into the `HashSet`.
5. Display the `HashSet`.
6. Use `contains()` to check whether an element exists.
7. Use `remove()` to remove an element.
8. Use `size()` to find the number of elements.
9. Use `isEmpty()` to check whether the `HashSet` is empty.
10. Use `clear()` to remove all elements.
11. Display the results.
12. Stop the program.

UML Diagram



+ remove(Object)	
+ contains(Object)	
+ size()	
+ isEmpty()	
+ clear()	

Code:

```
package Observations;
```

```
import java.util.HashSet;
```

```
public class HashSetClass {
```

```
public static void main(String[] args) {
```

```
    // Creating a HashSet
```

```
    HashSet<String> set = new HashSet<>();
```

```
    // add()
```

```
    set.add("Apple");
```

```
    set.add("Banana");
```

```
    set.add("Mango");
```

```
    set.add("Orange");
```

```
    set.add("Apple"); // Duplicate is not added
```

```
    System.out.println("HashSet after add(): " + set);
```

```
    // contains()
```

```
    System.out.println("Contains Banana: " + set.contains("Banana"));
```

```
// remove()

set.remove("Mango");

System.out.println("HashSet after remove(): " + set);


// size()

System.out.println("Size of HashSet: " + set.size());


// isEmpty()

System.out.println("Is HashSet empty? " + set.isEmpty());


// clear()

set.clear();

System.out.println("HashSet after clear(): " + set);


// isEmpty() after clear

System.out.println("Is HashSet empty after clear()? " + set.isEmpty());
}
}
```

Output:

```
es' '-cp' 'C:\Users\Sahithi pyla\AppData\Roaming\Code\User\w  
bservations.HashSetClass'  
HashSet after add(): [Apple, Mango, Orange, Banana]  
Contains Banana: true  
HashSet after remove(): [Apple, Orange, Banana]  
Size of HashSet: 3  
Is HashSet empty? false  
HashSet after clear(): []  
Is HashSet empty after clear()? true  
PS C:\Users\Sahithi pyla\Desktop\Sahithi_java> █
```

Result:

Thus, the Java program successfully demonstrates the HashSet methods add(), remove(), contains(), size(), isEmpty(), and clear().